Total No. of Questions : 10]

SEAT No. :

P3848

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B.E. (Computer Engineering)

PRINCIPLES OF MODERN COMPILER DESIGN

(2012 Course) (410442) (Semester - I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Draw and explain model of LR parser. **[4]**

b) Find FIRST and FOLLOW of the following : **[6]**

$S \rightarrow v \mid XUW \mid UV \mid uWV$

$U \rightarrow w \mid WvX \mid SUVW \mid SWX \mid \varepsilon$

$V \rightarrow x \mid wX \mid \varepsilon$

$W \rightarrow w \mid UXW$

$X \rightarrow yz \mid SuW \mid SW$

OR

Q2) a) Generate annotated parse tree for following expression : a*b–c/e + f. **[4]**

b) Construct LALR items for following grammar : **[6]**

$S \rightarrow AA,\ A \rightarrow aA|b.$

Q3) a) What is top down parsing? What are the problems with top down parsing? **[4]**

b) Write the quadruple, triple, indirect triple for the statement **[6]**

a: = b * –c + b * – c.

OR

Q4) a) Write short note on type checking & type conversion. **[4]**

b) Generate three address code for following code, **[6]**

While (a – b) do

If(cod) then

x = y + 2

else

x = y – 2.

P.T.O.

Q5) a) Write a note on a simple code generator. **[4]**

b) What are the principle sources of optimization? Give example of each. **[6]**

c) Write 3-Address code for following code and then perform optimization technique **[8]**

```
for (i = 0; i <= n; i++)
        {
          for (j = 0; j < n; i++)
             {
               c[i, j] = 0;
             }
        }
for (i = 0; i <= n; i++)
      {
        for (j = 0; j <= n; i++)
          {
            for (k = 0; k <= n; k++)
              {
                c[i, j] = c[i, j] + a[i, k]*b[k, j];
              }
          }
      }
```

OR

Q6) a) Define induction variable. Give examples. **[4]**

b) Explain the use of algebraic transformations with an example. **[6]**

c) Optimize and develop this code by eliminating common sub expression, performing reduction in strength on induction variables and eliminating all the induction variable you can **[8]**

```
t6 = 4*i                B2 : i = i + 1
x = a[t6]                    t2 = 4*i
t7 = 4*I                     t3 = a[t2]
t8 = 4*j                if t3 < a[t1] goto B2
t9 = a[t8]
a[t7] = t9
t10 = 4*j
a[t10] = x
goto B2.
```

Q7) a) Explain identification w.r.t. context handling. **[6]**

b) What are different types of routine. Explain each. **[6]**

c) List and Explain the Key features of functional programming language. **[4]**

OR

Q8) a) Explain Object Oriented Source language Issues. **[6]**

b) Write short note on Java CC. **[6]**

c) Give which aspect of Haskell corresponds to which compiler phase. **[4]**

Q9) a) Explain message passing and its issues related to parallel programming model. **[6]**

b) Describe the object replication, migration and location w.r.t. to Object Oriented languages. **[6]**

c) Give details of Just In Time compiler. **[4]**

OR

Q10) a) Differentiate between : **[6]**

i) Parallel and distributed systems.

ii) Multicomputers and multiprocessors.

b) Describe the compilation framework LLVM. **[6]**

c) What is interpreter? Explain JVM as an interpreter. **[4]**

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